CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Memory Organization

Lecture No.- 3



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Recap of Previous Lecture



- **Topic** Memory Address Decoder
- **Topic** Main Memory
- **Topic** RAM & ROM Chip
- **Topic** Multiple Chips in a Memory

Topics to be Covered



Topic Multiple Chips in Single Memory System

Topic DRAM Refresh

Topic Associative Memory



Topic : Multiple Chips in Single Memory System

2 chips of size 32×8 bits $\Rightarrow$ Total capacity $= 64 \times 8$ bits

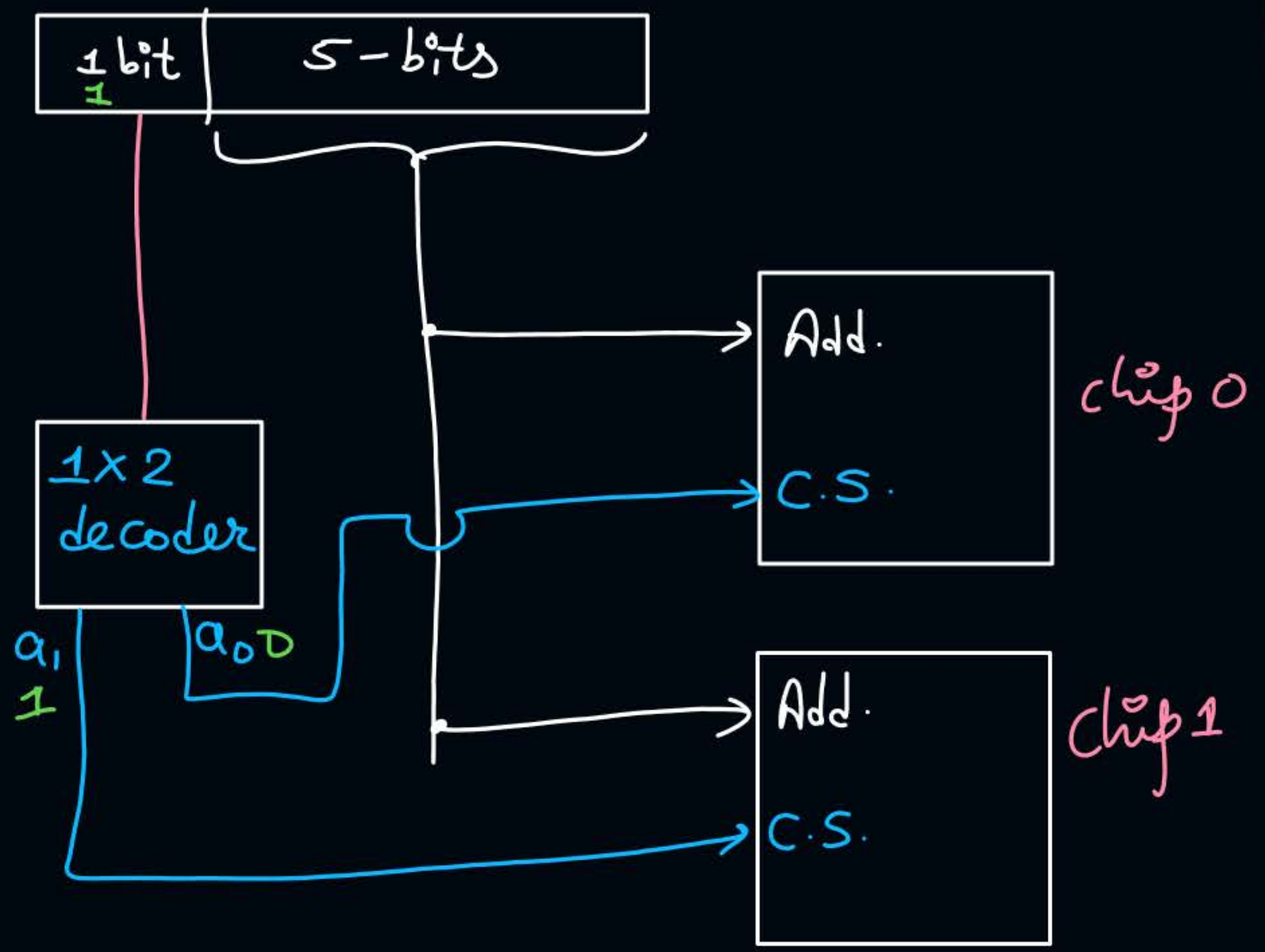
Address
$32 \times 8 \Rightarrow 5$ bits
$32 \times 8 \Rightarrow 5$ bits
$64 \times 8 \Rightarrow$ 6 bits

CPU generates

6 bits address range

0	0 0 0 0 0 0	chip 0
31	0 1 1 1 1 1	
32	1 0 0 0 0 0	chip 1
63	1 1 1 1 1 1	

CPU generates 6-bits add.



vertical arrangement of the chips.

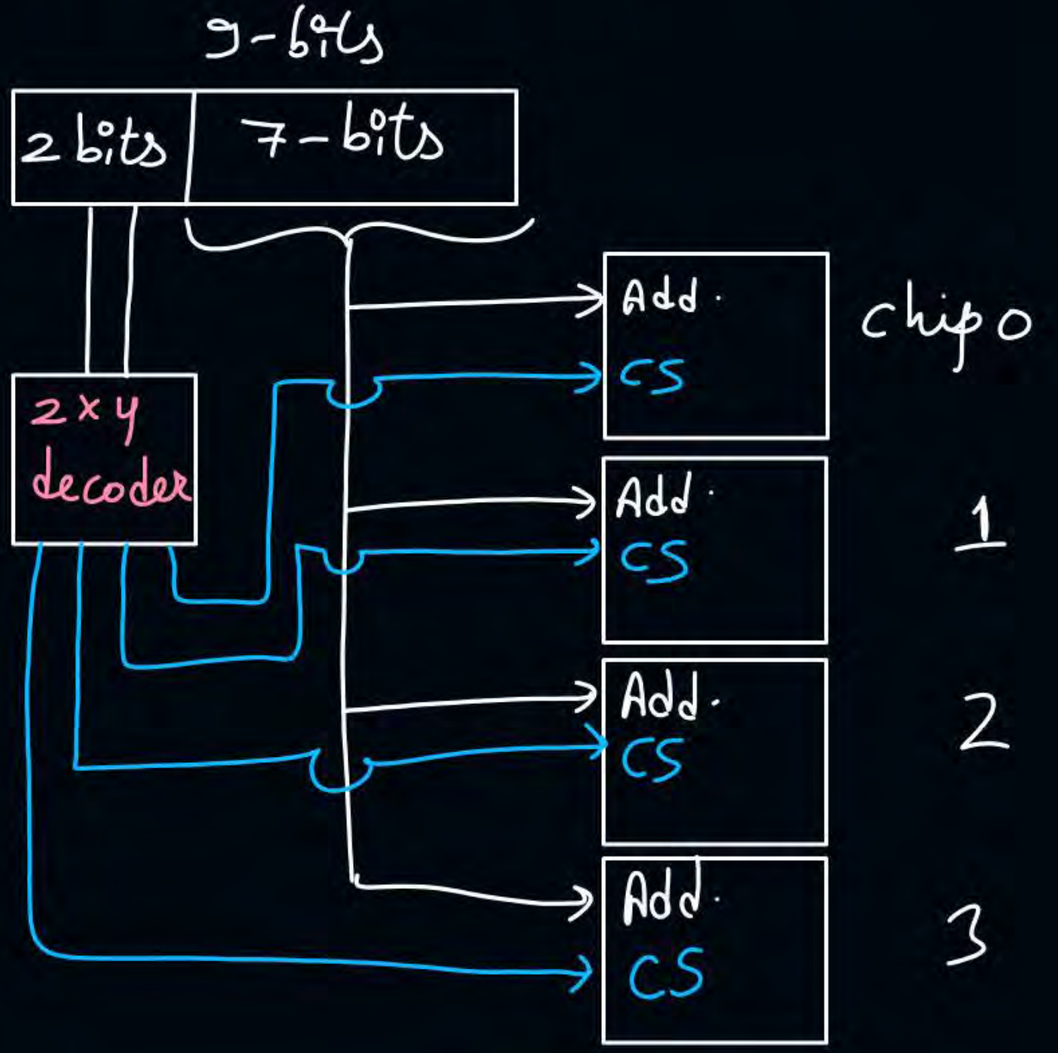
⇓

when no. of addresses needed in target mem. is more than no. of add. in a chip

ex:- (2)

4 chips of size 128×8 bits ⇒ Total capacity = 512 B

CPU generates add.



[NAT]

#Q. (a) How many 128×8 bits RAM chips are needed to provide a memory capacity of 2048 bytes?

(b) How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips?

(c) How many lines must be decoded for chip select? Specify the size of decoder?

a) No. of chips $= \frac{2048 \times 8 \text{ bits}}{128 \times 8 \text{ bits}} = \frac{2^{11}}{2^{7}} = 2^{4} = 16$

b) 11 bits/line,
7 lines common to all chips

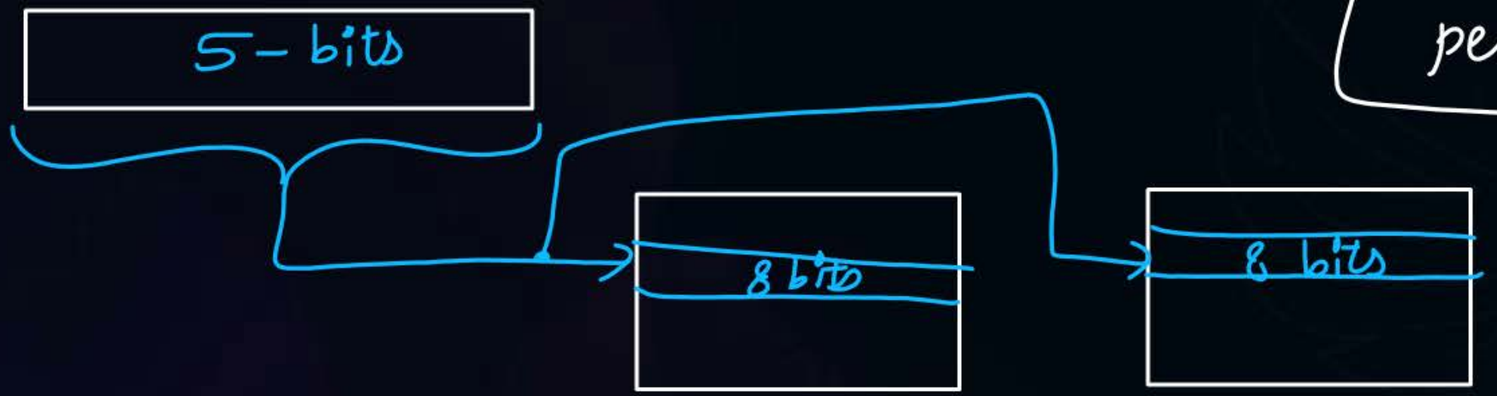
c) 4 lines, 4×16 decoder

[NAT]

#Q. How many 32×8 bits RAM chips are needed to provide a memory capacity of 32 ×16 bits?

$$\text{no. of chips} = \frac{32 \times 16}{32 \times 8} = 2 \text{ chips}$$

CPU generates 5 bits add.



Horizontal arrangement

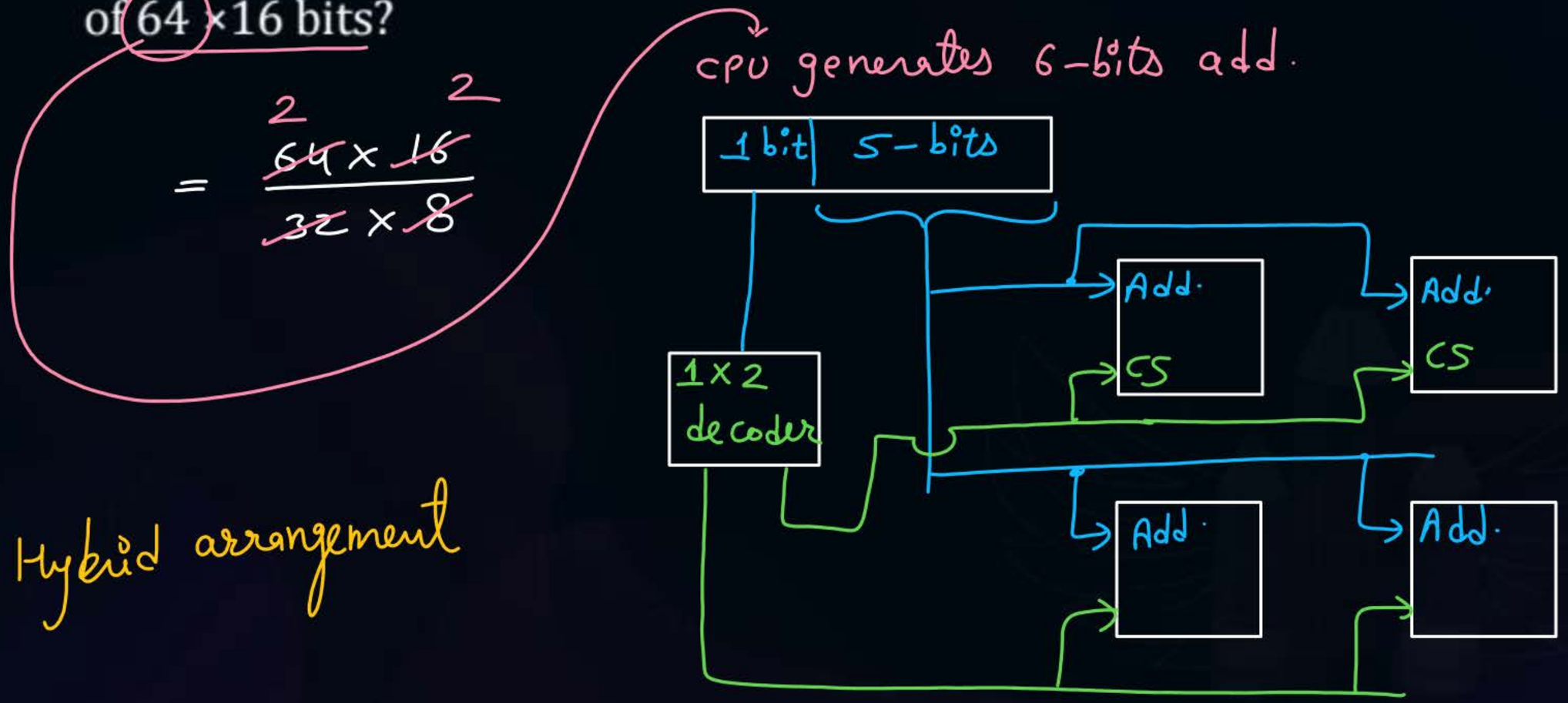
⇓

when data address is needed more than data per add. in chip

[NAT]

#Q. How many 32×8 bits RAM chips are needed to provide a memory capacity of 64 ×16 bits?

$$= \frac{64 \times 16}{32 \times 8}$$



Hybrid arrangement

[MCQ]

GATE- 2009

#Q. How many 32K × 1 RAM chips are needed to provide a memory capacity of 256K bytes?

default storage unit ⇒ bit

$$\frac{256KB}{32K \times 1} \quad \begin{matrix} 8 & 8 \end{matrix}$$

$= 64$

A. 8

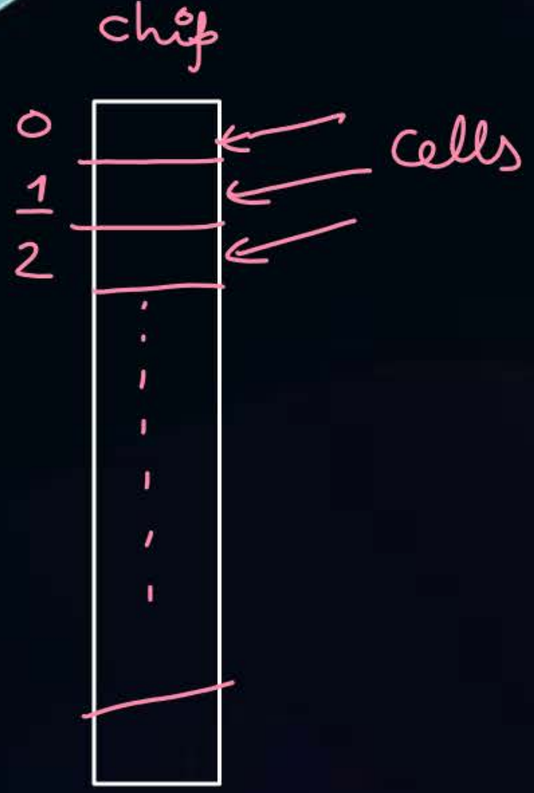
B. 32

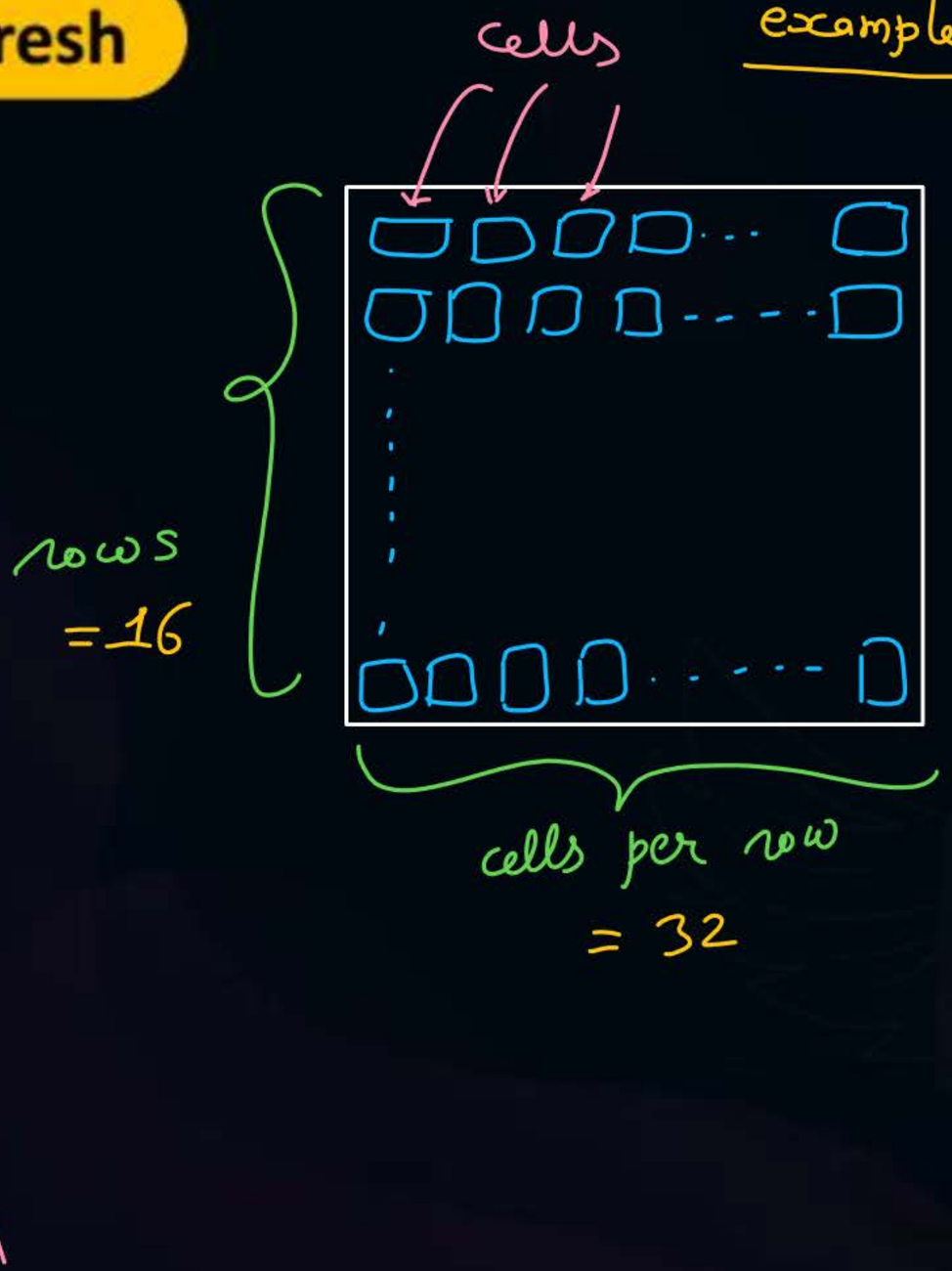
C. 64 ✓

D. 128



Topic : DRAM Refresh



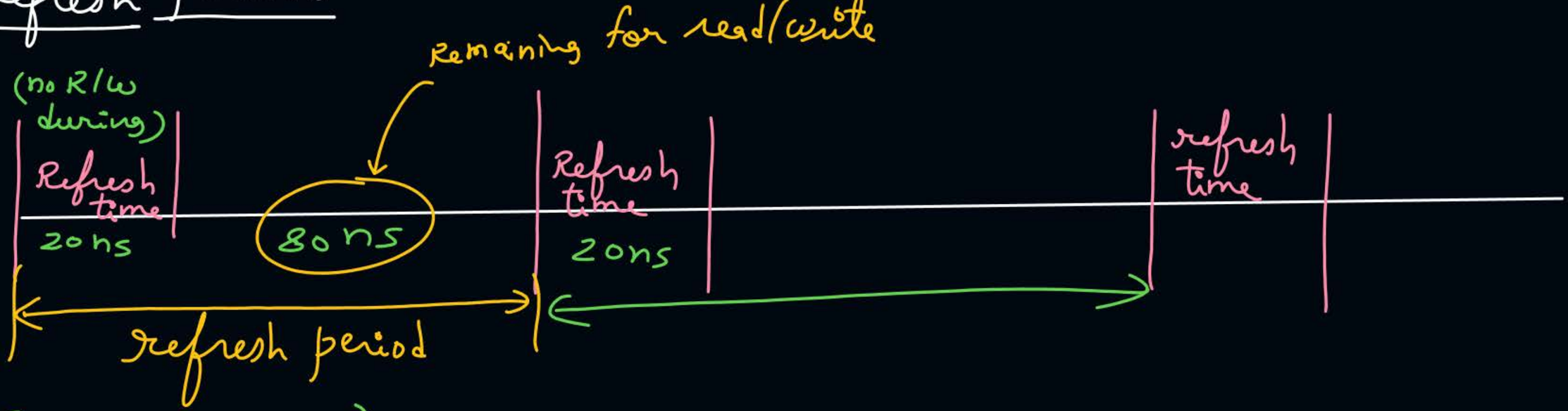


In one refresh operation, all cells of a row are refreshed.

1 chip refresh time = no. of rows of cells * 1 refresh operation time

n chips —"— —— = 1 chip refresh time

Refresh period :-



(assume 100 ns)

[NAT]

#Q. Consider a DRAM which can be refreshed in 10ns. The refresh period is 0.05 microseconds. = 50 ns

1. % of time taken in refresh? $\frac{10}{50} * 100\% = 20\%$
2. % of time remaining for read write is? $\frac{50-10}{50} * 100\% = 80\%$

[MCQ] GATE- 2010

#Q. A main memory unit with a capacity of 4 megabytes is built using 1M × 1-bit DRAM chips. Each DRAM chip has 1K rows of cells with 1K cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is?

A. 100 nanoseconds

B. 100×2^{10} nanoseconds ✓

C. 100×2^{20} nanoseconds

D. 3200×2^{20} nanoseconds

$= 1K * 100\,ns$

$\text{no. of chips} = \frac{4M * 8\ bits}{1M \times 1\ bits} = 32\ chips$



[NAT]

H.W.

#Q. A 32–bit wide main memory unit with a capacity of 1 GB is built using 256M X 4-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closet integer) of the time available for performing the memory read/write operations in the main memory unit is _____?

[NAT]

#Q. A DRAM chip of $256K \times 8$ bits has x rows of cells with y cells in each row? If DRAM takes 20ns for 1 refresh and 10.24 microseconds for entire chip refresh then the value of x + y is (1024)

Total chip refresh time = no. of rows * 1 refresh time

$$10.24\ \mu s = x * 20ns$$

$$x = \frac{10.24 * 10^{-6}}{20 * 10^{-9}}$$

$$= 0.512 * 1000$$

$$= 512$$

$$256k = 512 * y$$

$$y = \frac{2^{18}}{2^9} = 2^9 = 512$$

$$x + y = 512 + 512 = 1024$$



Topic : Associative Memory

faster
expensive

Used to implement
cache,
TLB

Known as content addressable memory also

→ No addresses of cells

→ searching is done based on content stored in cells.



faster than
SRAM



2 mins Summary

Topic Multiple Chips in Single Memory System

Topic DRAM Refresh

Topic Associative Memory



Happy Learning

THANK - YOU